Week 2 – OFDM Sync Short

This week's work focuses on the *OFDM Sync Short* block and its role in detecting IEEE 802.11a frames. The detection process exploits the periodic structure of the Short Training Field (STF), which consists of ten repetitions of a 16-sample sequence [1].

The receiver uses this periodicity to calculate the autocorrelation between samples separated by 16 positions. For a window of N_{win} samples, the autocorrelation can be expressed as

$$a[n] = \sum_{k=0}^{N_{\text{win}}-1} s[n+k]\, s^*[n+k+16], \tag{1}$$

where $s[n]$ denotes the received complex baseband signal and $(\cdot)^*$ denotes complex conjugation. The reference implementation uses a window of 48 samples [1].

The autocorrelation is normalized by the signal power:

$$p[n] = \sum_{k=0}^{N_{\text{win}}-1} |s[n+k]|^2, \qquad c[n] = \frac{|a[n]|}{p[n]}. \tag{2}$$

The normalized coefficient provides an indication of the periodic structure in the received signal and is used by the detection stage.

The *OFDM Sync Short* block analyses the normalized autocorrelation using a predefined threshold T. In the algorithm described in the reference paper, a detection plateau is identified when three consecutive coefficient values exceed the threshold:

$$c[n] > T, \qquad c[n+1] > T, \qquad c[n+2] > T. \tag{3}$$

Following detection, the block controls the forwarding of a fixed number of received samples to the subsequent processing stages. The block therefore performs frame-detection and stream-control functions; it does not decode the transmitted payload.

The practical work is based on a GNU Radio implementation of the receiver. The autocorrelation and power branches are implemented using delay, complex-conjugation, multiplication, magnitude, and FIR-filter operations. Both FIR filters are configured with 48 unit-valued taps and a decimation factor of one, implementing the moving-sum operations required for the autocorrelation and signal-power calculations.

Figure 1 shows the current GNU Radio Companion implementation, including the two FIR filters with their updated configurations.

An additional part of the work investigates the effect of changing the detection threshold. A threshold that is too low may increase false detections, whereas a threshold that is too high may cause valid frames to be missed. The limitations of the detection mechanism are also considered. In particular, the reference paper discusses the possibility of missing a subsequent frame while a fixed number of samples from a previously detected frame is being forwarded [1].

The experimental evaluation of different threshold values will be completed after the GNU Radio flowgraph has been successfully executed. No experimental results are reported at this stage.

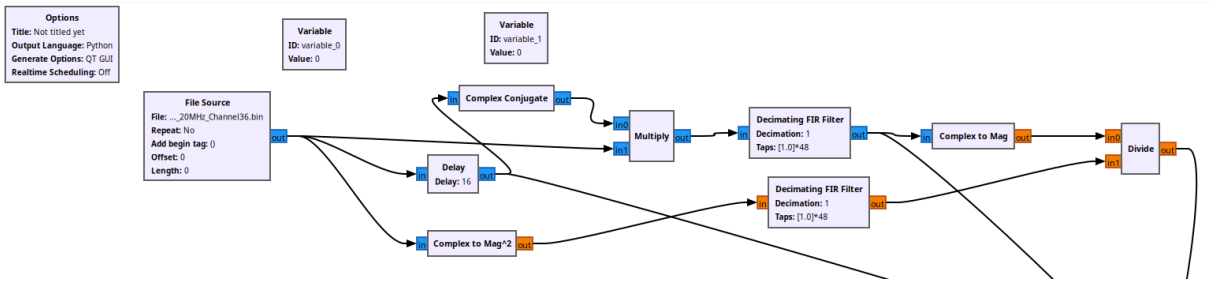


Figure 1: GNU Radio Companion implementation of the autocorrelation and signal-power branches, showing the updated 48-tap FIR filter configurations.

References

[1] B. Bloessl, M. Segata, C. Sommer, and F. Dressler, "An IEEE 802.11a/g/p OFDM Receiver for GNU Radio," in *Proceedings of the 2nd Workshop on Software Radio Implementation Forum (SRIF)*, 2013.